Corrected artifact publication: Setec Astronomy

The exact prepared twist follow-up archive is now published in `s3://setec-astronomy-artifacts/store`. An immutable conditional upload, native store registration, fresh-cache download and remote verification all succeeded. The earlier access denials used the wrong repository-default bucket, `crypto-autoresearcher`. They did not establish a permissions problem with the intended destination. The proposed policy for that other bucket is superseded and was never applied.

Requested scope and observed result

The user corrected the destination to Astronomy artifacts. Read-only bucket discovery returned the exact name `setec-astronomy-artifacts`. Publication used the existing prepared archive, with no repacking or scientific changes.

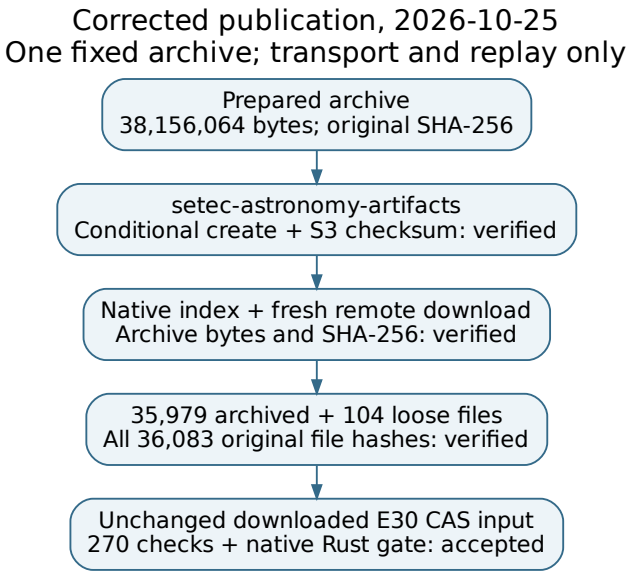
Requirement	Observed result
Correct destination	`setec-astronomy-artifacts`, with the existing `store/sha256/` layout
Immutable upload	Conditional PutObject succeeded; S3 returned the expected SHA-256 checksum
Native registration	Native push registered the exact name, digest and 38,156,064-byte size
Fresh remote download	New empty download cache; downloaded bytes and SHA-256 match
Remote verification	Native store reports cache OK and remote OK
Archived-member integrity	All 35,979 archived files match the original manifest
Complete tree reconstruction	All 104 separately tracked metadata files restored and checked; 36,083 files total
Frozen downloaded-input replay	Original 270-cell E30 incidence input passes PARI replay and the Rust gate
Scope of replay	Same host, one unchanged frozen input; full scientific rerun and independent-host replay remain open

Archive SHA-256:

`6fb0de2cf9ad5c2b9acd1703372560550433da7c095f5f42202100aba31ef167`

Logical artifact name:

`evidence/twist_routes_1_3_20261021/execution-v2-20261023/runs.tar.xz`



Execution and evidence

The conditional write uses `If-None-Match: *` and SHA-256 verification. The subsequent native push reports `uploaded: false` because that immutable object already exists after the successful conditional write; the combined receipt records the actual successful write. No object or existing name was replaced. No IAM or bucket policy was changed.

The study now has an explicit store config and native index. Reproduction sets both store selectors from the repository root. This keeps this artifact bound to the intended bucket without redirecting unrelated index entries.

The fresh download was extracted in a separate staging tree. Every file is hashed against the original full manifest, including the 104 loose metadata files restored from the checkout. Unexpected extracted files are rejected. The original E30 CAS script from that tree is byte-identical to the frozen 2026-10-23 input. Its PARI executable is also byte-identical to the prior run. The original Rust gate accepts the output marker, exit status, script/output hashes and error checks. All 270 original incidence checks replay; no new fixture, target, solver comparison or scientific experiment was introduced.

- Destination discovery
- Upload, native registration, fresh pull and remote verification
- Complete member-hash check
- Fresh CAS transport receipt
- Native CAS adjudication
- Current raw manifest
- Validation logs

Transport receipts retain exact nonsecret arguments, binary/source hashes, output hashes and UTC intervals. Credentials entered through non-echoing stdin and were passed only in child environments. No credential values, unrelated bucket names or caller identity values are persisted. The inherited native store binary's exact compiled source revision remains unknown; its binary hash is retained. This uncertainty does not change the checked byte identity.

The replay uses the unchanged native experiment binary from source revision:

`644f1dbef70c5935db211a7d3a70ec7bb0601336`

Its bounds are a 600-second cap, 4 GiB address space and one worker at isolation L0. No wall time is admitted as performance evidence. The editable diagram, rendered diagram and inspected PDF accompany this report under the repository's research-visuals guidance.

Correction and remaining requirements

The earlier denied attempts and wrong-bucket policy are preserved as historical evidence. Their access diagnosis is superseded. No change to that bucket's permissions is requested.

Remote publication and exact archive recovery are verified. A full rerun of every original scientific checker is not claimed. All sixteen research cards retain their blocked, unfunded states and their original mathematical/cost requirements. E26--E28 remain unrun. No calibrated IC speedup, novel attack, NIST logarithm or independent-host result is established.

The twelve previously committed oversized Git blobs still need a permitted owner/reviewer resolution, and exact-head CI and review remain required. Tracked remaining hold. The IC leaderboard, progress timeline and scientific geometry/energy figures are unchanged because this correction changes artifact availability only.